Tracking LQR with Riccati and Feed-forward Equations (Kirk Examples 5.2-3 and 5.2-4)

Optimal Control, Homework 5

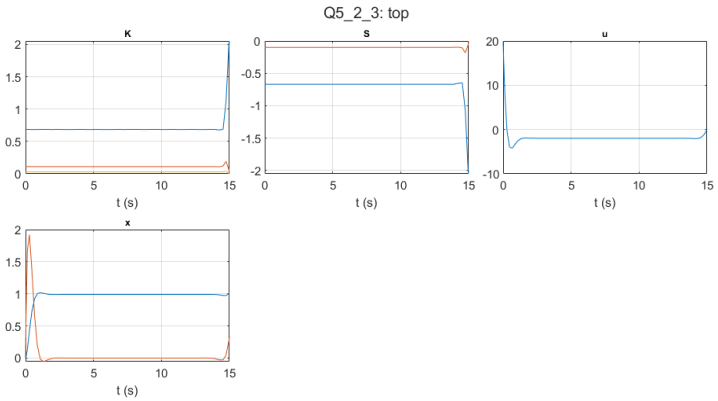
Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Spring 1404 (2025)

English summary of my original Persian report, which is kept as a PDF in this folder.

Both examples are linear tracking problems on the same second-order plant. The optimal control is $u^\star(t) = -R^{-1}B^T[K(t)x(t) + s(t)]$. Besides the Riccati equation for $K(t)$, a second linear equation for the feed-forward vector $s(t)$ is integrated backwards from the final time.

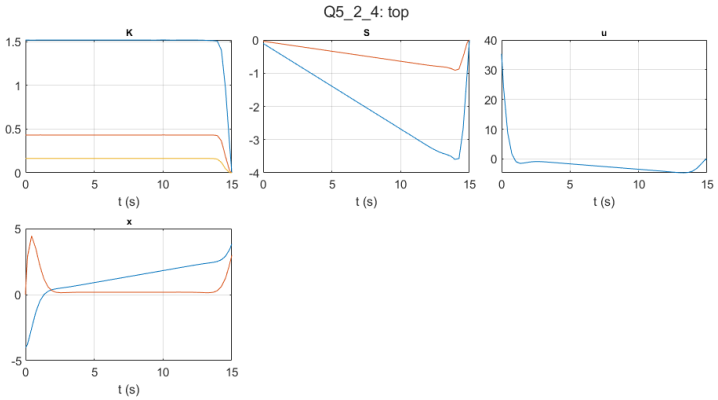
Example 5.2-3 (`Q5_2_3.slx`)

`Q5_2_3_Gains.m` integrates five equations (K_{11}, K_{12}, K_{22}, s_1, s_2) backwards from $t_f = 15\,\text{s}$ with `ode45`, using the terminal values from the book. The control law is in a MATLAB Function block (`myfcn`). My report notes that the curves match the book.



Example 5.2-4 (`Q5_2_4.slx`)

This case has a time-varying reference: the s_1 equation has a $0.4t$ forcing term and zero terminal conditions. The structure of the model is the same.



How to run: run the matching `*_Gains.m` script first, then the model.